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Smaller Cornu Ammonis (CA3) as a potential risk factor for suicidal behavior in mood disorders

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Abstract

Mood disorders and suicidal behavior have moderate heritability and familial transmission, and are associated with smaller hippocampal volumes. However, it is unclear whether hippocampal alterations reflect heritable risk or epigenetic effects of childhood adversity, compensatory mechanisms, illness-related changes, or treatment effects. We sought to separate the relationships of hippocampal substructure volumes to mood disorder, suicidal behavior, and risk and resilience to both by examining high familial risk individuals (HR) who have passed the age of greatest risk for psychopathology onset. Structural brain imaging and hippocampal substructure segmentation quantified Cornu Ammonis (CA1–4), dentate gyrus, and subiculum gray matter volumes in healthy volunteers (HV, N=25) and three groups with one or more relatives reporting early-onset mood disorder and suicide attempt: 1. unaffected HR (N=20); 2. HR with lifetime mood disorder and no suicide attempt (HR-MOOD, N=25); and 3. HR with lifetime mood disorder and a

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previous suicide attempt (HR-MOOD+SA, N=18). Findings were tested in an independent cohort not selected for family history (HV, N=47; MOOD, N=44; and MOOD+SA, N=21). Lower CA3 volume was found in HR (vs. HV), but not in MOOD and MOOD+SA (vs. HV), suggesting the finding reflects a familial biological risk marker, not illness or treatment-related sequelae, of suicidal behavior and mood disorder. Familial suicide risk may be mediated in part by smaller CA3 volume. The structure may serve as a risk indicator and therapeutic target for suicide prevention strategies in high-risk families.

Keywords

hippocampus subfields; risk factors; suicidal behavior; familial transmission; early life adversity; heritable

Introduction

Smaller hippocampal volume is a well replicated finding in brain imaging studies of mood disorder (Haukvik et al., 2022; Roddy et al., 2019; Sheline et al., 2002, 2019). Not all studies have found smaller hippocampal volumes in mood disorders (Greenberg et al., 2008; Phillips et al., 2015; Rusch et al., 2001; Videbech & Ravnkilde, 2004). The finding has been linked to illness progression, including longer illness duration (Bludau et al., 2016; Sheline et al., 1996), non-remission/treatment resistance (MacQueen et al., 2008; Phillips et al., 2015), and having more major depressive episodes (MacQueen et al., 2003). Preclinical and human research suggests that chronic stress and stress-mediated neurotoxic processes, including inflammation, hypothalamic-pituitary-adrenal axis dysfunction, and neurotransmitter disturbances, contribute to hippocampal volume loss as illness progresses (Belleau et al., 2019). Hippocampal volume differences disappeared when controlling for psychiatric comorbidities (Gosnell et al., 2020), suggesting hippocampal volume reductions are not specific to mood disorders. More pronounced hippocampal reductions in depressed suicide attempters have been observed relative to depressed non-attempters (Colle et al., 2015; Gosnell et al., 2016; Moica et al., 2016) suggesting these alterations are in part due to suicidal behavior in the context of mood disorder.

Automated hippocampal subfield segmentation revealed smaller volumes in bilateral Cornu Ammonis 1 (CA1), CA4, the dentate granule cell layer, molecular layer, and whole hippocampus in major depressive disorder (Han et al., 2019). Another study found that recurrent, as opposed to first-episode depression, had more pronounced volume deficits in left Cornu Ammonis (CA1-CA4), dentate gyrus, and subiculum, and left CA1 volume correlated with illness duration (Roddy et al., 2019).

There is debate about whether hippocampal volume reductions observed in mood disorders are a consequence of stress, mood disorder, and/or suicidal behavior, or whether they predate symptom onset and thereby contribute causally to pathophysiology (Belleau et al., 2019; Sheline et al., 2019). Previous studies have reported smaller hippocampus volumes in individuals at high familial risk for mood disorder (Amico et al., 2011; Chen et al., 2010). However, it is unknown whether hippocampus and/or hippocampal subfield deficits are associated with familial risk and pathogenesis for suicidal behavior and differ from the

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pathogenesis of mood disorder. It is also unknown whether hippocampal subfield deficits present prior to illness onset, or whether the deficits result after illness onset as part of downstream stress responses or damage due to factors like allostatic stress cortisol load, inflammation or excessive glutamate exposure (Kantrowitz et al., 2021).

To test whether hippocampal subfield alterations are associated with risk for developing suicidal behavior in the context of mood disorder, or whether they are related to compensatory adaptation, resilience, or illness-related changes, the current study examined regional differences in hippocampal subfield volumes in two separate cohorts, one in which participants were selected for familial risk of suicidal behavior and mood disorder ("FAMPATH"), and another in which participants were not selected for this familial risk ("second dataset"). Both cohorts included a group of low familial risk healthy volunteers (HV). The FAMPATH cohort included medication-free high familial risk individuals, having a first or second degree relative with both an early onset mood disorder and a history of suicidal behavior. High risk individuals were either unaffected (i.e. HR, recruited when largely beyond the age of risk for mood and suicidal behavior), or affected with mood disorder alone (HR-MOOD) or mood disorder plus one or more lifetime suicide attempts (HR-MOOD+SA).

We conducted 3 pairwise group contrasts comparing 8 hippocampal subfield volumes between HV and each high risk group (HR, HR-MOOD, HR-MOOD+SA). We hypothesized the strongest differences would consist of lower CA1–4 volumes in HR-MOOD+SA (vs. HV) based on previous research (Colle et al., 2015; Gosnell et al., 2016; Moica et al., 2016; Roddy et al., 2019). The HR vs. HV comparison was performed to test for alterations in these regions that might suggest markers of familial risk or resilience (compensatory adaptation), but not illness-related changes. We tested whether findings from the primary FAMPATH dataset could be replicated in the second dataset that was not selected for family history, and further explored the influence of participants' familial risk status on replicability in a small subset of participants that were defined as "HR-participants" based on similar selection criteria as used in the FAMPATH dataset.

Methods

Subjects

The FAMPATH sample consisted of high familial risk individuals and healthy volunteers (HV) that were 25 years of age or older, an age by which mood disorder and suicidal behavior in our familial transmission studies are generally phenotypically clinically expressed (Brent et al., 2015; Melhem et al., 2019). All high-risk subjects were categorized as being at-risk for a mood disorder and suicide because they had a first or second degree relative with early-onset (<40 years) mood disorder and a lifetime history of suicide attempt. The subjects in the high-risk group were separated into vulnerable and resilient groups based on their phenotype: individuals with a lifetime history (current or past) of mood disorders and a history of suicide attempt (HR-MOOD+SA, n=18); individuals with a lifetime history (current or past) of mood disorders but no suicide attempt (HR-MOOD, n=25); and resilient (HR, n=20) individuals with no personal history of mood disorder or suicide attempt. Qualifying mood disorders included major depressive disorder, bipolar disorder, dysthymia

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and depressive disorder not otherwise specified (NOS). The HR-MOOD sample consisted of 20 participants with major depressive disorder, 1 with bipolar disorder, and 4 participants with depressive disorder NOS. The HR-MOOD+SA sample consisted of 14 participants with major depressive disorder, 3 with bipolar disorder, and 1 with depressive disorder NOS. Healthy volunteers (HVs, n=35) had no personal or family history of psychiatric disorders or suicidal behavior. Additional exclusion criteria for all groups included current alcohol and substance use disorder, other Axis I psychiatric diagnosis (except current or lifetime comorbid anxiety disorder, lifetime alcohol and substance use disorder, post-traumatic stress disorder), and family history of schizophrenia. See Supplementary Methods for assessments and additional inclusion/exclusion criteria for FAMPATH and for the second independent cohort that was not selected for family history. The sample was recruited at two sites, University of Pittsburgh Medical Center and New York State Psychiatric Institute, in protocols approved by the respective Institutional Review Boards (IRB) at the University of Pittsburgh and The New York State Psychiatric Institute. Participants were recruited through outpatient clinics, clinician referrals and IRB-approved advertisements and all participants provided written, informed consent. See Melhem et al. 2022 for more information regarding comorbid conditions in each group and differences between study sites (Melhem et al., 2022). See Box 1 for a recapitulation of the above abbreviations for reference throughout the manuscript.

Final sample sizes across two independent cohorts

The primary discovery dataset (FAMPATH) consisted of 25 HV, 20 HR, 25 HR-MOOD, and 18 HR-MOOD+SA (Table 1, top half). The second dataset included 47 HV, 44 individuals with current mood disorder (MOOD), and 21 participants with MOOD and one or more lifetime suicide attempts (MOOD+SA, see Table 1, bottom half). The second dataset included 8 individuals who were considered high-familial risk on the basis of having one or more first degree relatives with mood disorder and one or more first degree relatives with a previous suicide attempt (i.e. “HR-participants”). The second dataset (see Supplementary Table S1) consisted of data collected and analyzed the same way as was the FAMPATH cohort in terms of brain imaging acquisition and segmentation analysis. Our group has not previously published on hippocampal volumetric MRI findings in either cohort.

T1 structural brain image acquisition and preprocessing

The FAMPATH data were collected from two sites: Pittsburgh and New York (NY), while the second dataset was collected from only the NY site. The Pittsburgh and NY sites had standardized MRI protocols. MRI images were acquired on a GE 3 Tesla MR750 scanner utilizing 18 high order superconducting shim coils and 5 resistive shim coils, and acquiring images using a 32-channel head-coil at the New York site and on a Siemens 3T Trios scanner in Pittsburgh. A sagittal T1-weighted structural scan was acquired over the whole brain with the following parameters: repetition time (TR), 6 ms; echo time (TE), 2.4 ms; flip angle, 8°; 1-mm slice thickness (zero gap); 178 slices; and field of view, 256 × 256mm. All images were reconstructed to a size of 256 × 256, with an isotropic resolution of $1mm^3$. The structural images were first processed using recon-all with FreeSurfer 7.1 (Fischl & Dale, 2000). Details of the recon-all and Freesurfer segmentation process are described elsewhere (Fischl et al., 2004; Salat et al., 2004). Next the hippocampus subfields were segmented

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using the segmentHA_T1.sh bash script (Iglesias et al., 2015). Residual plots and histograms of resulting volumes were inspected for outliers.

Linear Mixed Effects modeling

Matlab table structures were used to store hippocampus and eTIV volumes and clinical (group) labels, and then write the data to csv files for subsequent analyses. Linear mixed effects by Restricted Maximum Likelihood with t-tests using Satterthwaite's method (lmerModLmerTest) models were applied using R Version 3.6.3. Analysis with log volume as outcome focused on 3 main models/comparisons: 1) HV vs. HR-MOOD+SA; 2) HV vs. HR-MOOD; and 3) HV vs. HR. Since we anticipated that group differences would vary across subfields (Roddy et al., 2019), the primary models included the fixed effects region, age, sex, log of total intracranial volume, site, and group x region interaction. Subject was treated as a random effect. A second set of models included the same terms as the primary models with the exception that the group x region interaction term was replaced with the main effect of group.

In the first (FAMPATH) dataset, the first two models (HV vs. HR-MOOD+SA and HV vs. HR-MOOD) were used to identify subfields altered in mood disorder (HR-MOOD) and suicide attempt in the context of mood disorder (HR-MOOD+SA) in high familial risk individuals, and the third model (HV vs. HR) was used to identify putative markers of familial risk or resilience that were not due to illness-related changes. This analysis (and all preceding analyses) include all 8 hippocampal subfields in order to "borrow strength" across regions within each subject. The subfields found to differ in this first dataset were considered as ROIs in the comparisons in the second independent dataset that was not selected for family history. Uncorrected p-values are reported for transparency. False discovery rate (FDR) correction was applied across the 8 regions in each model and asterisks in resulting tables indicate which regions survive $p<0.05$ corrected. Results surviving a more liberal threshold of $p<0.05$ uncorrected are also briefly discussed in the context of previous findings.

Results

Demographic and clinical results

Within both the primary FAMPATH and second dataset, the subgroups did not differ in sex ratio or age (all p-values > 0.2, Table 1). The HR-MOOD and HR-MOOD+SA groups in the FAMPATH dataset had lower current Beck Depression Inventory (BDI) and Hamilton Depression Rating Scale (HAM17) scores since the study included individuals with lifetime diagnosis of mood disorder, but they were not required to be currently ill (Table 1). The MOOD and MOOD+SA subgroups in the second dataset had higher BDI and HAM17 scores because it involved currently depressed subjects (see Table 1 and Supplementary Table S1).

The FAMPATH HR, HR-MOOD, and HR-MOOD+SA groups were selected to have at least one or more 1st or 2nd degree relative with early onset MOOD+SA (<40 yrs). The participants in the second dataset were not selected for family history of MOOD+SA. We

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compared the number of participants with one or more 1st degree relatives with SA or MOOD in the MOOD and MOOD+SA subsamples of each dataset (Table 1). Thirty-five of 43 participants (81%) in the FAMPATH HR-MOOD and HR-MOOD+SA subsamples had at least one 1st degree relative with MOOD, and 30 of 43 participants (70%) had at least one 1st degree relative with SA. In the combined MOOD and MOOD+SA subsamples of the second dataset, 31 of 65 participants (48%, vs. FAMPATH Fisher's exact test p-value=0.0005) had a 1st degree relative with MOOD, and 8 of 65 participants (12%, vs. FAMPATH Fisher's exact test p-value <0.0001) had a 1st degree relative with SA. There were eight MOOD+SA or MOOD individuals in the second dataset that met similar selection criteria for affected HR-participants in the FAMPATH sample (i.e. one or more 1st degree relatives with lifetime MOOD and one or more 1st degree relatives with lifetime SA).

Both depressed suicide attempters and unaffected individuals with high familial suicide risk have smaller CA1 and CA3 volumes compared to healthy volunteers

The first model comparing HV vs. HR-MOOD+SA groups in the FAMPATH dataset found no evidence for an overall region x group interaction ($F_{7,134}$=1.32, p=0.24) indicating that group differences did not differ significantly by region. T-tests using Satterthwaite's method suggested lower volumes in CA1 body (p=0.02 uncorrected) and CA3 body (p=0.01 uncorrected) in HR-MOOD+SA relative to HV (gray shaded rows in Table 2) but no region survived correction for multiple comparisons. The model which included a group term instead of a group x region interaction term suggested trend level evidence for overall lower hippocampal subfield volumes in HR-MOOD+SA vs. HV ($F_{1,47}$=4.0, p=0.05).

The model comparing HV to HR-MOOD found no evidence for region x group interactions and no individual regions survived p<0.05 uncorrected (Table 3). In addition, there was no main effect detected in the HR-MOOD vs. HV comparison that included a group term instead of a group x region interaction term ($F_{1,44}$=0.79, p=0.38).

The model comparing HV to HR found no evidence for overall region x group interaction ($F_{7,141}$=1.44, p=0.19). T-tests (with a focus on regions identified in the first model) found revealed lower CA3 body volume (p=0.0057 uncorrected, p<0.05 FDR corrected) in HR relative to HV (gray shaded rows in Table 4) and trend-level evidence suggesting lower left CA1 body volumes in HR vs. HV (p=0.067 uncorrected). Similar regions that were smaller in HR-MOOD+SA vs. HV (CA1 body and CA3 body) tended to be smaller in HR relative to HV, though only the lower CA3 body volume in HR vs. HV comparison survived multiple comparisons correction. Finally, the model that included a group term instead of a group x region interaction term revealed trend-level evidence for lower volumes across all regions in HR vs. HV ($F_{1,39}$=3.54, p=0.07).

Depressed suicide attempters and non-attempters not selected for familial risk do not show smaller CA1 body and CA3 body relative to healthy volunteers

Similar models in the two studies comprising the second dataset found no region x group interactions in MOOD+SA vs. HV, while there was trend-level evidence for *greater* CA1 body volumes (t(169)=1.9, p=0.06) and *greater* CA3 body volume (t(71)=2.8, p=0.006 in MOOD vs. HV (see Supplementary Table S2).

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Post hoc exploratory analysis of hippocampus subregion volumes in the second dataset using a subsample selected for familial risk

The second dataset was not selected for family history (i.e. MOOD and MOOD+SA participants were not required to have one or more 1st degree or 2nd degree relatives with lifetime MOOD+SA). We defined a subset of participants as "HR" if they had one or more 1st degree relatives with lifetime MOOD and one or more 1st degree relatives with lifetime SA (N=8 "HR-participants"). We hypothesized these individuals will tend to have lower CA1 and CA3 volumes relative to non-HR participants (MOOD and MOOD+SA). To explore this, the CA1 and CA3 body volumes were submitted to a Mann-Whitney U test. Consistent with our hypothesis, the HR-participants had lower left CA1 and C3 body volumes than the non-HR participants, though neither region reached statistical significance (HR-participants < non-HR participants and HV; left CA1 body two-tailed p=0.12 uncorrected; left CA3 body two-tailed p=0.21 uncorrected, Supplementary Figure 2).

Testing relationships between hippocampal subfield volumes and illness duration and severity

We tested whether the number of lifetime major depressive episodes was correlated (Spearman's r) with left CA1 and CA3 body volumes (raw values or adjusted for age, sex and eTIV) in all participants with available data (HR-MOOD, HR-MOOD+SA, MOOD and MOOD+SA, total n=75). For left CA3 adjusted for age, sex and TIV: r=−0.06, p=0.60; left CA3 raw values: r=−0.059, p=0.62; for left CA1 adjusted for age, sex and TIV: r=0.06, p=0.59, left CA1 raw values: r=0.09, p=0.45, data not shown).

Discussion

In this study we find evidence for smaller hippocampal CA3 volume in participants with **high familial risk for** early onset mood disorder and suicide attempt, but who themselves have no lifetime mood disorder diagnosis or suicide attempt and who have passed through the age of greatest risk for psychopathology onset. This finding is consistent with related studies that have found smaller hippocampus volumes in healthy individuals at-risk for depression (Amico et al., 2011; Chen et al., 2010). Previous studies have also found lower hippocampal volumes in depressed suicide attempters (vs. depressed non-attempters) (Colle et al., 2015; Gosnell et al., 2016; Moica et al., 2016; Sarkinaite et al., 2021) as well as lower left CA1–4 volumes in recurrent depression and greater illness duration (Haukvik et al., 2022; Roddy et al., 2019). Here, we sought to replicate their results in our data and find that high familial risk individuals with lifetime mood disorder and previous suicide attempt had smaller CA3 and CA1 volumes relative to HV, though the findings did not survive correction for multiple comparisons. Moreover, similar differences were not observed in our second replication dataset that was not selected for family history. Taken together, the above findings suggest low CA3 volume may be a biological risk factor for suicidal behavior in the context of mood disorder. That the high familial risk individuals did not develop depression and suicidal behavior (despite having smaller CA3 volumes) suggests the absence of other risk factors (i.e. environment, chronic stress, etc.) (Belleau et al., 2019) and/or the undetected presence of neural compensatory mechanisms involving other brain regions that explains the absence of psychopathology in this high familial risk group.

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Our results are consistent with a recent study that found hippocampal subfields (CA1–4, DG) have high heritability (h^2 ~ 0.75–0.85) (Whelan et al., 2016). The study analyzed young, healthy twins from the Queensland Twins Imaging Study (N=728) and found that of 12 hippocampal subfields examined, CA1 had the second highest heritability (h^2 = 0.84) while CA3 was near the median (h^2 = 0.75). By comparison the hippocampal fissure had relatively low (but still statistically significant) heritability (h^2 = 0.56). These results are also consistent with those from a large mega-analysis (N>20K) that found CA1–4 had higher heritability relative to other subfields using SNP-based heritability analysis (van der Meer et al., 2020).

In addition to heritability (i.e. genetic factors), early environmental factors such as early life trauma or stress may also comprise a familial risk factor for mood disorder and suicide that results in lower hippocampal volumes (Teicher et al., 2018; Vythilingam et al., 2002; Weissman et al., 2020; Woon & Hedges, 2008). Consistent with our findings, previous studies found smaller volumes specifically in CA1/3 or DG in children with early life adversity (Dahmen et al., 2018; Teicher et al., 2012, 2018). It is possible that genetics or early adversity, or both, contribute to lower CA1/3 volumes in individuals at higher familial risk for mood disorder and suicidal behavior.

Contrary to our expectations, we did not find low hippocampal subfield volumes in MOOD or MOOD+SA groups not selected for family history, nor did we observe an association between hippocampal CA1–4 volumes and number of previous major depressive episodes. This is consistent with some previous studies that found no differences in hippocampal volumes in mood disorder (Greenberg et al., 2008; Phillips et al., 2015; Rusch et al., 2001; Videbech & Ravnkilde, 2004). These null findings are consistent with the interpretation that low hippocampal volumes are associated with risk for psychopathology, but not illness severity and sequelae. This interpretation may also explain findings of lower CA1–4 and other principal hippocampus substructure volumes in participants with recurrent vs. first presentation depression and with longer symptom durations (Roddy et al., 2019), since these outcomes may reflect a higher risk for depression rather than effects of illness progression.

Our study is limited by a relatively small sample size. None of the regions from the models comparing HV with each patient group survived multiple comparisons correction, while only the CA3 body volume in the HR vs. HV comparison survived correction. Smaller CA1 and CA3 volumes in a small independent subset of participants selected (post hoc) for family history were observed in the second dataset, but findings did not reach statistical significance, possibly due to limited power. Additional research in independent cohorts is required to confirm and extend the findings. Future research should also examine the role of CA3 (and CA1) in emotional and cognitive functioning associated with psychopathology, which may reveal more sensitive neural markers of risk or resilience for mood disorders and suicidal behavior.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data availability statement

The derived data that support the findings of this study are available upon request from the corresponding authors. The raw data are not publicly available as participants of this study did not agree for their data to be shared publicly.

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Box 1.

Abbreviations used to describe groups throughout the manuscript.

HV=healthy volunteers. Individuals who have no lifetime Axis I disorder diagnosis and no 1st or 2nd degree relative with mood disorder or previous suicide attempt; **MOOD**=participants with lifetime mood disorder, not selected for family history; **SA**=suicide attempt; **MOOD+SA**=participants with lifetime mood disorder and history of suicide attempt, not selected for family history; **HR**=unaffected, high familial risk "resilient" individuals. Individuals >25 years of age with a 1st or 2nd degree relative with early-onset MOOD+SA and who themselves have passed the age of greatest risk for psychopathology absent history of MOOD or MOOD+SA; **HR-MOOD**=affected, high familial risk "non-resilient" individuals. Individuals >25 years with a 1st or 2nd degree relative with MOOD+SA and who themselves have lifetime history of early-onset MOOD; **HR-MOOD+SA**=affected, high familial risk "non-resilient" individuals. Individuals >25 years of age with a 1st or 2nd degree relative with MOOD+SA and who themselves have lifetime early onset MOOD+SA;

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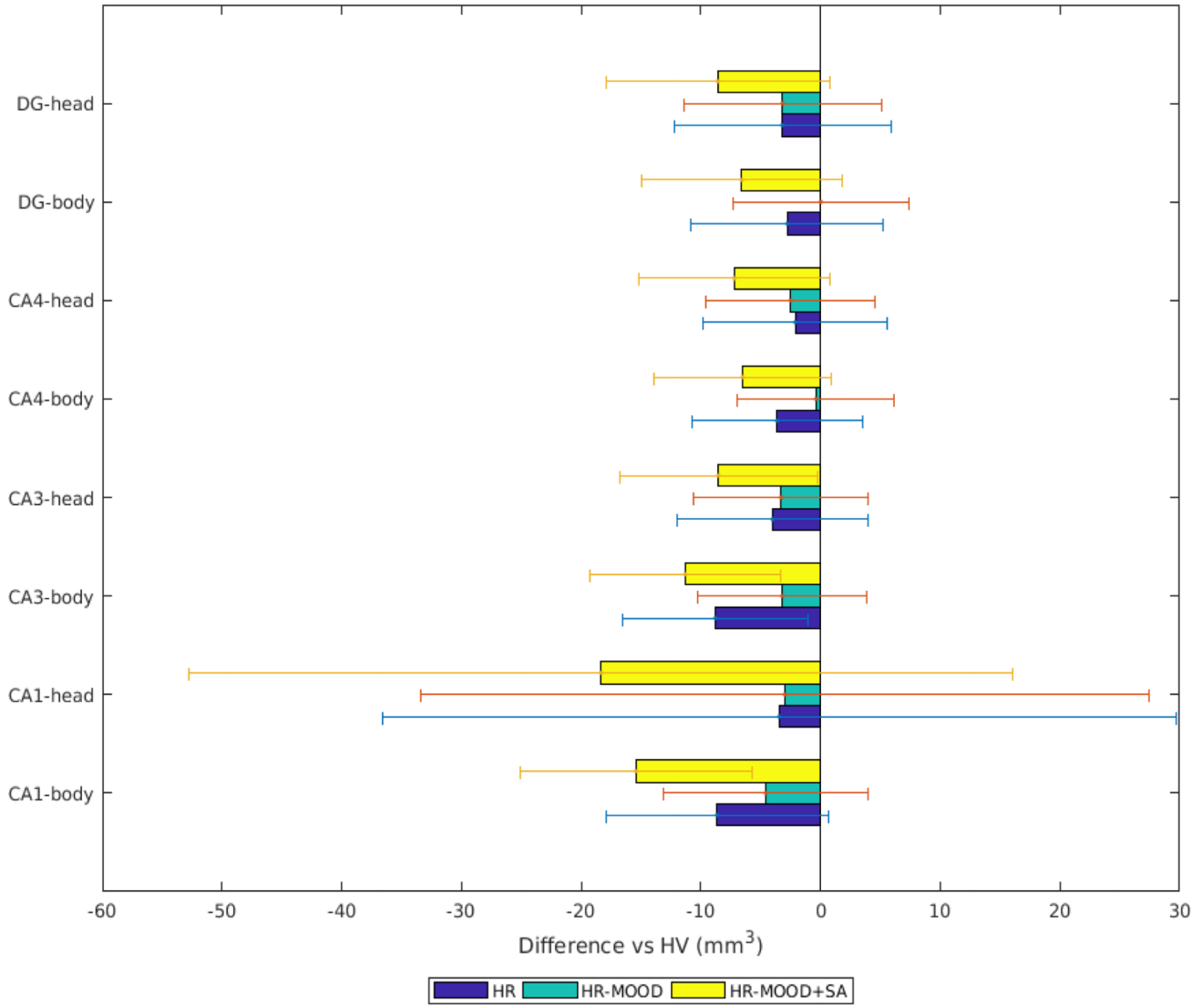

Figure 1. Mean volume differences of left hippocampal subfields in the FAMPATH dataset.

Bar plots represent mean volume differences relative to the HV group in HR (blue), HR-MOOD (gray), and HR-MOOD+SA (yellow) groups. Error bars represented 2*standard error. **HV**=healthy volunteers, **HR**=unaffected high-familial risk individuals, **HR-MOOD**=affected high-familial risk individuals with early onset mood disorder, and **HR-MOOD+SA**=affected high-familial risk individuals with early onset mood disorder and suicide attempt (see Box 1).

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Table 1.

Demographics and behavioral data.

FAMPATH	**HV (N=25)**	**HR (N=20)**	**HR-MOOD (N=25)**	**HR-MOOD+SA (N=18)**	**pval**
Age: mean (iqr)	40.24 (15.25)	38.85 (20.50)	36.24 (15.50)	38.17 (18.00)	0.56
HAM17: mean (iqr)	0.88 (1.25)	1.30 (2.00)	3.56 (4.75)	9.61 (12.00)	<0.001
BDI: mean (iqr)	1.24 (1.25)	2.05 (4.00)	6.72 (8.50)	16.39 (8.00)	<0.001
eTIV: mean (iqr)	1415283.80 (255923.50)	1334939.75 (157165.97)	1415271.33 (222531.34)	1317348.85 (192865.95)	0.06
Sex: count (%)	8/17 (0.32/0.68)	5/15 (0.25/0.75)	9/16 (0.36/0.64)	6/12 (0.33/0.67)	0.89
FamDep: count (%)	25/0 (1.00/0.00) 5/15 (0.25/0.75)		4/21 (0.16/0.84)	4/14 (0.22/0.78)	<0.001
FamSui: count (%)	25/0 (1.00/0.00) 5/15 (0.25/0.75)		5/20 (0.20/0.80)	8/10 (0.44/0.56)	<0.001
site: count (%)	15/10 (0.60/0.40) 7/13 (0.35/0.65)		11/14 (0.44/0.56)	8/10 (0.44/0.56)	0.39
Second dataset	**HV (N=47)**	**NA**	**MOOD (N=44)**	**MOOD+SA (N=21)**	**pval**
Age: mean (iqr)	32.91 (14.25)	NA	34.55 (11.50)	28.76 (13.50)	0.06
HAM17: mean (iqr)	0.94 (1.00)	NA	17.55 (6.00)	17.19 (9.00)	<0.001
BDI: mean (iqr)	0.89 (0.75)	NA	24.82 (10.00)	22.85 (8.00)	<0.001
eTIV: mean (iqr)	1394551.18 (244377.45)	NA	1392891.96 (207518.54)	1325280.01 (170449.62)	0.29
Sex: count (%)	17/30/0 (0.36/0.64/0.00)	NA	15/28/1 (0.34/0.64/0.02)	9/12/0 (0.43/0.57/0.00)	0.74
FamDep: count (%)	47/0 (1.00/0.00)	NA	22/22 (0.50/0.50)	12/9 (0.57/0.43)	<0.001
FamSui: count (%)	47/0 (1.00/0.00)	NA	40/4 (0.91/0.09)	17/4 (0.81/0.19)	0.02
site: count (%)	35/6/6 (0.74/0.13/0.13)	NA	26/9/9 (0.59/0.20/0.20)	20/0/1 (0.95/0.00/0.05)	0.05

Group differences were assessed using Kruskal–Wallis one-way ANOVA (continuous variables) or chi^2 test (categorical variables). Measures of familial risk are indicated by FamDep and FamSui where No/Yes=0/(1 or more) 1st degree relatives with MOOD or suicide attempt (SA), respectively. iqr=interquartile range. **HV**=healthy volunteers; **HR**=unaffected high-familial risk individuals; **HR-MOOD**=affected high-familial risk individuals with early onset mood disorder; **HR-MOOD+SA**=affected high-familial risk individuals with early onset mood disorder and suicide attempt; **MOOD**=individuals with mood disorder; **MOOD+SA**=individuals with mood disorder and previous suicide attempt (see Box 1); **BDI**=Beck Depression Inventory; **HAM17**=17 item Hamilton Depression Rating Scale.

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Table 2.

Results from linear mixed effects model comparing 8 hippocampal subfields in HV vs. HR-MOOD+SA in FAMPATH.

Grey rows indicate regions surviving p<0.05 uncorrected. No regions survived correction for multiple comparisons.

Fixed Effects	Estimate	Std. Error	df	t-value	Pr(>|t|)
regionCA1 (body)	−2.361	2.416	37.003	−0.977	0.335
regionCA1 (head)	−0.913	2.416	37.003	−0.378	0.708
regionCA3 (body)	−2.775	2.416	37.003	−1.148	0.258
regionCA3 (head)	−2.398	2.416	37.003	−0.993	0.327
regionCA4 (body)	−2.387	2.416	37.003	−0.988	0.330
regionCA4 (head)	−2.356	2.416	37.003	−0.975	0.336
regionGC-ML-DG (body)	−2.255	2.416	37.003	−0.933	0.357
regionGC-ML-DG (head)	−2.155	2.416	37.003	−0.892	0.378
age	−0.001	0.002	37.000	−0.462	0.647
sex	−0.023	0.038	37.000	−0.615	0.542
log(etiv)	0.506	0.170	37.000	2.971	0.005
site	−0.031	0.033	37.000	−0.933	0.357
regionCA1 (body):groupHV	0.098	0.042	70.218	2.348	0.022
regionCA1 (head):groupHV	0.042	0.042	70.218	1.008	0.317
regionCA3 (body):groupHV	0.110	0.042	70.218	2.65	0.010
regionCA3 (head):groupHV	0.060	0.042	70.218	1.449	0.152
regionCA4 (body):groupHV	0.061	0.042	70.218	1.468	0.147
regionCA4 (head):groupHV	0.065	0.042	70.218	1.562	0.123
regionGC-ML-DG (body):groupHV	0.057	0.042	70.218	1.376	0.173
regionGC-ML-DG (head):groupHV	0.058	0.042	70.218	1.407	0.164

HV=healthy volunteers; **HR-MOOD+SA**=affected high-familial risk individuals with early onset mood disorder and suicide attempt (see Box 1). **etiv**=estimated total intracranial volume; **CA**=cornu ammonis; **GC-ML-DG**=granule cell layer of dentate gyrus and molecular layer; **body**=main body of the hippocampus; **head**=anterior to the body; **sex**=female vs. male difference; **site**=site 2 vs. site 1 difference; **:groupHV**=HV vs. HR-MOOD+SA difference.

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Table 3.

Results from linear mixed effects model comparing 8 hippocampal subfields in HV vs. HR-MOOD in FAMPATH.

No regions differed between groups at p<0.05 uncorrected.

Fixed Effects	Estimate	Std. Error	df	t-value	Pr(>|t|)
regionCA1 (body)	−1.304	1.849	44.010	−0.705	0.485
regionCA1 (head)	0.118	1.849	44.010	0.064	0.950
regionCA3 (body)	−1.707	1.849	44.010	−0.923	0.361
regionCA3 (head)	−1.355	1.849	44.010	−0.733	0.468
regionCA4 (body)	−1.335	1.849	44.010	−0.722	0.474
regionCA4 (head)	−1.310	1.849	44.010	−0.709	0.482
regionGC-ML-DG (body)	−1.202	1.849	44.010	−0.65	0.519
regionGC-ML-DG (head)	−1.116	1.849	44.010	−0.603	0.549
age	−0.001	0.001	44.000	−0.396	0.694
sex	−0.018	0.030	44.000	−0.608	0.547
log(etiv)	0.433	0.130	44.000	3.342	0.002
site	−0.001	0.025	44.000	−0.056	0.956
regionCA1 (body):groupHV	0.040	0.033	131.400	1.195	0.234
regionCA1 (head):groupHV	0.011	0.033	131.400	0.322	0.748
regionCA3 (body):groupHV	0.042	0.033	131.400	1.247	0.215
regionCA3 (head):groupHV	0.017	0.033	131.400	0.511	0.610
regionCA4 (body):groupHV	0.009	0.033	131.400	0.262	0.794
regionCA4 (head):groupHV	0.019	0.033	131.400	0.576	0.566
regionGC-ML-DG (body):groupHV	0.004	0.033	131.400	0.126	0.900
regionGC-ML-DG (head):groupHV	0.019	0.033	131.400	0.568	0.571

HV=healthy volunteers; **HR-MOOD**=affected high-familial risk individuals with early onset mood disorder (see Box 1). **etiv**=estimated total intracranial volume; **CA**=cornu ammonis; **GC-ML-DG**=granule cell layer of dentate gyrus and molecular layer; **body**=main body of the hippocampus; **head**=anterior to the body; **sex**=female vs. male difference; **site**=site 2 vs. site 1 difference; **:groupHV**=HV vs. HR-MOOD difference.

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Table 4.

Results from linear mixed effects model comparing 8 hippocampal subfields in HV vs. HR in FAMPATH.

Grey row indicates the CA3 body is smaller in HR relative to HV (p=0.006 uncorrected, p<0.05 corrected).

Fixed Effects	Estimate	Std. Error	df	t-value	Pr(>|t|)
regionCA1 (body)	−1.336	1.789	39.006	−0.747	0.460
regionCA1 (head)	0.092	1.789	39.006	0.052	0.959
regionCA3 (body)	−1.770	1.789	39.006	−0.989	0.329
regionCA3 (head)	−1.385	1.789	39.006	−0.774	0.443
regionCA4 (body)	−1.373	1.789	39.006	−0.768	0.447
regionCA4 (head)	−1.334	1.789	39.006	−0.746	0.460
regionGC-ML-DG (body)	−1.238	1.789	39.006	−0.692	0.493
regionGC-ML-DG (head)	−1.142	1.789	39.006	−0.638	0.527
age	−0.002	0.001	39.000	−2.14	0.039
sex	−0.078	0.029	39.000	−2.751	0.009
log(etiv)	0.442	0.126	39.000	3.502	0.001
site	0.013	0.025	39.000	0.524	0.604
regionCA1 (body):groupHV	0.062	0.033	100.444	1.857	0.066
regionCA1 (head):groupHV	0.025	0.033	100.444	0.764	0.447
regionCA3 (body):groupHV	0.094	0.033	100.444	2.826	0.006*
regionCA3 (head):groupHV	0.037	0.033	100.444	1.102	0.273
regionCA4 (body):groupHV	0.037	0.033	100.444	1.1	0.274
regionCA4 (head):groupHV	0.033	0.033	100.444	0.991	0.324
regionGC-ML-DG (body):groupHV	0.030	0.033	100.444	0.895	0.373
regionGC-ML-DG (head):groupHV	0.034	0.033	100.444	1.033	0.304

* = survives p<0.05 false discovery rate corrected over the 8 regions. **HV**=healthy volunteers; **HR**=unaffected high-familial risk individuals; **HR-MOOD+SA**=affected high-familial risk individuals with early onset mood disorder and suicide attempt (see Box 1). **etiv**=estimated total intracranial volume; **CA**=cornu ammonis; **GC-ML-DG**=granule cell layer of dentate gyrus and molecular layer; **body**=main body of the hippocampus; **head**=anterior to the body; **sex**=female vs. male difference; **site**=site 2 vs. site 1 difference; **:groupHV**=HV vs. HR difference.